

PERTH METRO, Australia

WMO: 946080

Lat: 31.9192S

Lon: 115.8728E

Elev: 83

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.9	41.2	33.1	27.7	63.9	35.4	30.4	59.5	18.8	60.5	16.4	59.7	0.9	50	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.5	97.5	68.3	94.0	67.6	90.7	67.1	72.8	86.2	71.1	85.4	69.8	83.6	8.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.8	106.2	77.7	67.1	100.0	76.0	65.6	94.8	74.9	36.5	85.1	35.0	85.4	33.8	83.5	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	15.0	13.7	34.7	104.9	1.7	3.0	33.5	107.0	32.5	108.8	31.6	110.5	30.3	112.7		
(5)			34.5	75.3	1.6	2.1	33.3	76.8	32.3	78.0	31.4	79.2	30.2	80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.3	75.8	76.0	73.1	67.2	61.2	56.9	55.5	56.3	58.4	62.7	68.4	72.4		
	DBStd	9.05	5.89	5.88	5.94	5.24	4.45	3.86	4.06	3.76	4.11	5.15	5.80	6.30		
	HDD50	8	0	0	0	0	0	2	5	1	0	0	0	0		
	HDD65	1329	1	1	6	34	131	245	296	271	204	108	26	6		
	CDD50	5577	800	729	715	517	348	207	175	196	251	393	551	695		
	CDD65	1421	336	310	255	101	14	1	0	1	4	36	127	236		
	CDH74	12963	3265	3062	2214	683	118	2	1	7	54	322	1093	2142		
	CDH80	5539	1514	1426	919	187	15	0	0	0	8	89	424	957		
(14) Wind	WSAvg	6.5	8.2	8.0	7.2	5.7	4.8	4.9	5.2	5.4	6.2	7.0	7.8	8.2		
(15) Precipitation	PrecAvg	32.3	0.3	0.7	0.7	1.8	4.4	7.0	6.7	4.5	3.0	2.0	0.9	0.4		
	PrecMax	41.0	2.0	5.2	2.8	3.9	8.7	11.6	10.7	7.6	7.2	5.0	2.9	1.3		
	PrecMin	22.0	0.0	0.0	0.0	0.0	0.6	2.7	3.5	0.7	0.7	0.0	0.1	0.0		
	PrecStd	5.4	0.4	1.1	0.7	1.1	2.0	2.5	1.8	1.5	1.5	1.1	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.5	101.1	99.3	90.5	81.9	73.4	71.2	74.3	79.9	88.1	96.0	100.5		
		MCWB	68.0	70.2	68.5	65.0	60.8	57.2	56.9	57.5	59.8	63.4	64.8	67.5		
	2%	DB	96.4	96.3	93.3	85.1	77.3	70.2	67.8	70.2	74.1	81.8	89.7	94.8		
		MCWB	68.2	69.0	67.6	63.7	60.5	57.6	57.0	58.0	59.0	61.9	63.3	66.4		
	5%	DB	92.5	92.6	88.9	81.2	73.7	67.8	65.6	67.2	70.3	77.0	84.6	90.0		
		MCWB	68.5	68.6	66.7	63.6	60.0	57.2	56.4	57.0	58.3	60.5	63.3	66.1		
	10%	DB	88.4	88.8	84.9	77.6	70.8	65.6	63.6	64.8	67.3	73.0	80.2	85.1		
		MCWB	67.6	68.0	65.8	63.3	59.5	56.8	55.8	56.3	57.4	59.8	62.6	65.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.3	75.6	72.9	69.5	66.8	63.4	62.2	63.1	63.5	66.4	68.9	71.7		
		MCDB	89.9	90.1	85.3	77.2	71.6	67.4	65.4	67.5	71.1	79.4	83.0	90.5		
	2%	WB	72.7	73.2	71.0	68.1	64.8	61.5	60.0	60.9	61.6	64.4	67.2	69.7		
		MCDB	86.4	85.9	82.9	76.3	70.4	65.9	63.8	65.9	69.1	75.5	80.7	86.0		
	5%	WB	70.9	71.3	69.5	66.6	63.1	59.8	58.4	59.2	60.1	62.8	65.7	68.3		
		MCDB	85.1	86.3	81.6	75.4	69.2	64.5	62.6	64.4	67.4	72.8	78.3	83.7		
	10%	WB	69.2	69.7	68.2	65.2	61.4	58.3	57.1	57.6	58.5	61.3	64.3	66.7		
		MCDB	83.1	84.1	80.1	74.2	68.1	63.1	61.6	62.9	65.5	70.6	76.4	80.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	21.5	20.8	19.8	19.7	18.2	17.1	17.5	17.7	18.9	19.5	19.9		
		MCDBR	28.1	27.4	26.9	24.9	24.7	21.6	19.6	22.4	23.5	25.8	28.0	28.3		
		MCWBR	9.7	9.3	9.0	9.9	11.4	11.5	11.0	11.9	11.4	11.1	10.3	9.8		
	5% WB	MCDBR	21.7	22.3	20.5	18.6	17.1	16.2	14.4	16.4	18.5	20.6	21.9	22.9		
		MCWBR	8.8	8.9	8.2	9.0	10.1	10.5	9.9	10.3	10.6	10.0	9.2	9.0		
(40) Clear-Sky Solar Irradiance	taub	0.340	0.341	0.325	0.318	0.308	0.301	0.300	0.308	0.327	0.327	0.329	0.329			
	taud	2.579	2.593	2.629	2.635	2.635	2.633	2.634	2.605	2.527	2.545	2.566	2.585			
	Ebn at Noon	317	311	305	291	278	272	278	290	299	311	318	321			
	Edh at Noon	34	32	29	26	23	22	23	26	31	33	34	34			
(44) All-Sky Solar Radiation	RadAvg	2599	2343	1935	1413	1055	861	912	1185	1584	2063	2458	2662			
	RadStd	120	80	95	82	69	59	57	58	91	89	103	109			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page